

Q3 PS8) $\frac{1}{e^x + 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1}$

Q $n = \frac{g}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2}{e^{E(p)/kT} \pm 1} dp$ | $n = \frac{N}{Vol}$

→ Similar to ISM: $\int \mathcal{E}(u) dM : \# / Vol$
 $\int \underline{M} \mathcal{E}(u) dM : Mass / Vol$

$n = \frac{g}{(2\pi\hbar)^3} \int \frac{4\pi p^2 \mathcal{E}(p)}{e^{E(p)/kT} \pm 1} dp$

$n_B = \frac{g_B}{(2\pi\hbar)^3} \int \frac{4\pi p^2 \mathcal{E}(p)}{e^{E(p)/kT} - 1} dp$ | $n_F = \frac{g_F}{(2\pi\hbar)^3} \int \frac{4\pi p^2 \mathcal{E}(p)}{e^{E(p)/kT} + 1} dp$

* Rel. cases: $E(p) = pc$

$n = \frac{g}{(2\pi\hbar)^3} \int \frac{4\pi p^2 (pc)}{e^{pc/kT} \pm 1} dp$

* Trying to find a total $\sim n_B + n_F$

* $g_{\pm} = \sum g_B \mp \frac{7}{8} \sum g_F$

→ $\frac{1}{e^x + 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1}$

* $f_F(T) = f_B(T) - f_B(T/2)$

$n_B = \frac{g_B}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi c p^3}{e^{pc/kT} - 1} dp$

→ $y = pc/kT = \tau$ $dy = \left(\frac{c}{kT}\right) dp$

$p^3 = \left(\frac{kT}{c}\right)^3 y^3$ | $dp = \left(\frac{kT}{c}\right) dy$

$n_B = \frac{g_B}{(2\pi\hbar)^3} \int_0^\infty 4\pi c \left(\frac{kT}{c}\right)^3 y^3 \left(\frac{kT}{c}\right) dy$

$= \frac{g_B 4\pi c \left(\frac{kT}{c}\right)^4}{(2\pi\hbar)^3} \int_0^\infty \frac{y^3 dy}{e^y - 1}$

$\int_0^\infty \frac{y^3 dy}{e^y - 1} = \frac{\pi^4}{15}$

$n \sim (kT)^4$

$$u_B = \frac{4\pi}{(2\pi\hbar)^3} \frac{1}{c^3} (kT)^4 \frac{\pi^4}{90} 6 = \frac{g_B \pi^2}{30 (hc)^3} (kT)^4$$

* Also want num. $n_B = \left(\right) \int_0^\infty \frac{y^2}{e^y - 1} dy$

$$n_B = \frac{g_B}{\pi^2} \int_0^\infty \left(\frac{y^2}{hc} \right)^3$$

③ Want to show $u_F(T) = \frac{7}{8} \frac{g_F}{g_B} u_B(T)$
 similar for $n_F(T) = \frac{3}{4} \frac{g_F}{g_B} n_B(T)$

$$u_F = \frac{g_F}{(2\pi\hbar)^3} \int \frac{4\pi p^3 c dp}{e^{pc/kT} + 1}$$

$$\frac{1}{e^x + 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1}$$

$$= \frac{g_F}{(2\pi\hbar)^3} \left(\int \frac{p^3 dp}{e^{pc/kT} - 1} - \int \frac{2p^3 dp}{e^{2pc/kT} - 1} \right) \left(\frac{g_B}{g_B} \right)$$

$$\tilde{y} = \left(\frac{2pc}{kT} \right)^3 \quad d\tilde{y} = \frac{2pc}{kT}$$

$$= u_B(T) - \frac{2}{16} u_B = u_B - \frac{1}{8} u_B = \frac{7}{8} u_B(T)$$

$$u_F = \frac{7}{8} \left(\frac{g_F}{g_B} \right) u_B$$

$$S(T) \equiv \frac{4}{3} \frac{u}{T}$$

$$S_B = \frac{4}{3} \frac{u_B}{T}$$

$$S_F = \frac{4}{3} \frac{u_F}{T}$$

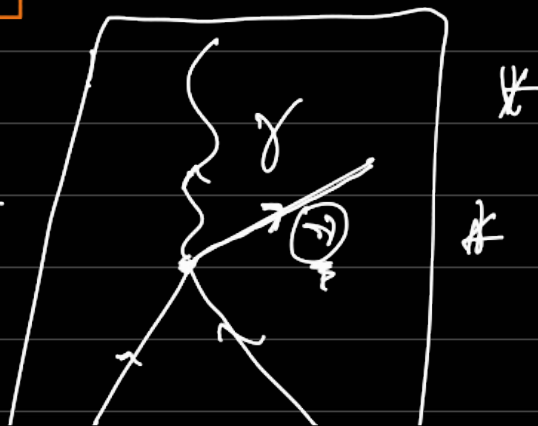
$$\frac{S_B}{S_F} = \frac{u_B}{u_F} = \frac{8}{7} \left(\frac{g_B}{g_F} \right)$$

$$S \sim \frac{(kT)^4}{T} \sim T^3$$

* $g_{e^+} = g_{e^-} = g_\nu = 2$

→ ν : neutrino

$u, 1/3$



want: $T(\gamma) = (11) T(\gamma)$

• $S_{\text{before}}(t'') = S_{\text{after}}(t')$

$S_{\text{after}}(\gamma) = S_{\text{before}}(e^+, e^-)$

→ we use the relationship above

$$\frac{S_b(\gamma)}{S_b(e^+, e^-)} = \frac{8}{7} \frac{g_\gamma}{(g_{e^+} + g_{e^-})}$$

$$S_b(e^+, e^-) = \frac{(g_{e^+} + g_{e^-})}{g_\gamma} \frac{7}{8} S_b(\gamma)$$

$$\begin{aligned} S_b(e^+ + e^- + \gamma) &= S(e^+, e^-) + S_b(\gamma) \\ &= \frac{(g_{e^+} + g_{e^-})}{g_\gamma} \frac{7}{8} S_b(\gamma) + S_b(\gamma) \\ &= \left(\frac{(g_{e^+} + g_{e^-})}{g_\gamma} \frac{7}{8} + 1 \right) S_b(\gamma) \end{aligned}$$

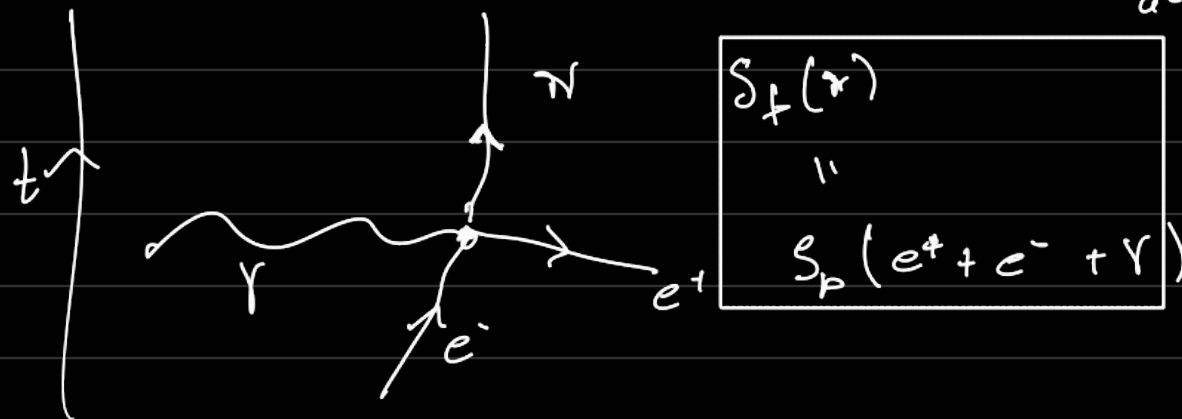
$g = 2$

$$\begin{aligned} &= \left(\left(\frac{4}{2} \right) \frac{7}{8} + 1 \right) S_b(\gamma) \\ &= \left(\frac{7}{4} + 1 \right) S_b(\gamma) \end{aligned}$$

$S_b(e^+ + e^- + \gamma) = \frac{11}{4} S_b(\gamma)$

From assuming S is conserved

$$\begin{aligned} S_b(e^+ + e^- + \gamma) &= S_a(\gamma) \end{aligned}$$



$$S_a(\gamma) = \frac{11}{4} S_b(\gamma)$$

$$T^3(\gamma) = \frac{11}{4} T(\gamma)^3 \Rightarrow T(\gamma) = \left(\frac{11}{4} \right)^{1/3} T(\gamma)$$

→ changes with redshift + cosmology

① (i) $d_A(z)$: Angular in cosmology

$\theta r = d_A$
 $\theta d_A = d \leftarrow \text{cosm.}$



$$d_A(z) = \frac{R(t) S_A(r)}{1+z}$$

$d_L(z)$: Luminosity distance

$$\text{Flux} = \frac{L}{4\pi r^2}$$

$$r \rightarrow d_L(z) = R(t) S_A(r) (1+z)$$

$$\uparrow \quad d_A(z) \quad \uparrow$$

$$\text{Flux} = \frac{L}{4\pi d_L(z)^2} = \frac{L}{4\pi R_0^2 S_A^2(r) (1+z)^2}$$

$$\frac{1}{(1+z)^2}$$

(ii) $S_\nu d\nu = \frac{L_\nu}{4\pi R_0^2 S_A^2(r) (1+z)^2} d\nu$

$$L = \frac{dE}{dt} \rightarrow (1+z) \frac{dE}{dt} = (1+z) L$$

$$\Delta E \rightarrow (1+z) dE$$

$$\Delta \lambda \rightarrow (1+z) d\lambda$$

(iii) $d\nu = d\left(\frac{c}{\lambda}\right) = -\frac{d\lambda}{\lambda^2} \rightarrow \frac{(1+z)}{\lambda^2} \rightarrow \frac{(1+z)}{(1+z)^2} = \frac{1}{(1+z)} d\lambda$

$$S_\nu = \frac{(1+z) L}{4\pi R_0^2 S_A^2 (1+z)^2} \left(\frac{1}{(1+z)} \right)$$

$$S_\nu = \frac{L_\nu(r)}{4\pi R_0^2 S_A^2 (1+z)^2}$$

(iv) $\nu \rightarrow (1+z) \nu$

(v) $\Delta \lambda : S_\nu \rightarrow S_\nu / (1+z)$

$$S(\nu) = S\left(\frac{c}{\lambda}\right) \Rightarrow S\left(\frac{c}{(1+z)\lambda}\right) \approx \frac{1}{(1+z)} S_\nu$$

$$\lambda \rightarrow (1+z) \lambda$$

$$S_\nu \rightarrow \frac{1}{(1+z)} S_\nu$$

$$\frac{1}{(1+z)} S_N(v_0) = \frac{L((1+z)v_0)}{4\pi R_0^2 S_u^2 (1+z)^2}$$

$$S_N(v_0) = \frac{L_N((1+z)v_0)}{4\pi R_0^2 S_u^2 (1+z)}$$

* Distance modules:

<u>A.1</u>	2.5	5
<u>A.2</u>	3.5	
<u>A.3</u>	3	



A.4 | 2 (cm: red)
H₀ $\Lambda_1 \rightarrow \Lambda_2$

3

A.5 | 2

A.6 | 4

k=0:

$$H^2 = \frac{8\pi G}{3} \rho_c \Rightarrow$$

$$\rho_c = \frac{3H^2}{8\pi G}$$

$$\Omega_i = \rho_i / \rho_c$$

A.7 | * Time

A.8 | 3.5

B.1 | (a) 2.5-3

(b) 3.5

(c) 3.5

(d) Cover tomorrow

B.2 | (a) 2

(b) 2-2.5

(c) $\rightarrow$ Plots from notes: look at is understand 2.5

① 3

② 4

* B.3] ① 3

② 2.5

$\Omega(t)$

③ 2

④ 2

$$H = \frac{1}{2t}$$

}

• A.5

• A.6

• A.7

• B.2 (a) ✓

• B.3

